

# CURRICULUM VITA

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## **Farn Wang, Professor**

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### **Working Experience in the Past:**

Associate professor, National Taiwan University, 2002-2005  
Associate Research Fellow, Academia Sinica, 1997-2002  
Assistant Research Fellow, Academia Sinica, 1993-1997

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### **Education:**

Ph.D. Computer Science, The University of Texas at Austin, 1993  
M.S., Computer Engineering, National Chiao-Tung University, 1984  
B.S., Electrical Engineering, National Taiwan University, 1982

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### **Tools Implemented:**

STW (Software Testing Workbench) 2007  
RED (A model-checker for timed automata and linear-hybrid automatas), 2002  
SGM (State-Graph Manipulator) with Bow-Ann Hsiung, 2002

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### **Tutorials:**

1. 7th ICFEM, Nov. 2005, Birmingham, UK
  2. 24th IFIP FORTE, September 27-30, 2004, Madrid, Spain,
  3. 1stATVA, Taipei, Taiwan, ROC, Dec. 2003
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### **Invited Speech:**

1. 1stSYNT, July 2012, Berkeley, California, USA.
  2. 1st IWTS (International Workshop on Timed Systems), March 3-5, 1999, Kyoto, Japan,
  3. 1998 IEEE HASE (High-Assurance Software Engineering), Nov. 13-14, Washington, D.C., USA
  4. 2ndAVIS (Automated Verification of Infinite-State Systems), April 12, 2003, Warsaw, Poland
  5. SVV (Software Verification & Validation) 2005, Oct. 2005, Birmingham, UK.
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### **Journal Editorships:**

1. FMDS (International Journal on Formal Methods for System Designs), Springer-Verlag(SCI extended).
  2. IJFCS ( International Journal on Foundations of Computer Science), special issue on infinite system verification and analysis, 2003 (SCI extended)
  3. IJFCS,special issue on ATVA 2003/2004 (SCI extended)
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**Book Editors:**

1. Proceedings of RTCSA'1997, IEEE Computer Society.
2. Proceedings of RTC'1999.
3. Proceedings of ATVA'2003.
4. Proceedings of ATVA'2004, in LNCS, Springer-Verlag
5. Proceedings of FORTE'2005, in LNCS, Springer-Verlag
6. ATVASteering committee member

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**Conference General Chair:**

1. IEEE TASE 2010
2. ICTAC 2016

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**Conference Program Chairs:**

1. 3rd ATVA, 2004, speakers include
2. IFIP 25'th FORTE, 2005, Keynote speakers
3. ICTAC 2016

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**Conference Program Co-chairs:**

1. ICTAC 2016,
2. RTCSA1997,
3. ATVA2003,
4. RTC 1999

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**Program Committee Memberships since 1996:**

1. APSDAC 2006, 2008
2. APSEC 2000, 2001, 2003, 2004
3. ASIAN 1999
4. ATVA 2003-2016
5. AVIS 2003
6. ECRTS 2003, 2004, 2005
7. EUC 2004
8. ICTAC 2016
9. IFIP FORTE2004, 2005, 2006, 2007
10. ICFEM 2002, 2003, 2004, 2005, 2006
11. IEEE RTAS1996, 1997, 2000, 2001, 2004, 2006
12. RTCSA 1996-2000, 2002-2005
13. IEEE RTSS1996, 1999, 2000, 2001, 2004
14. ICESS 2007
15. FMICS 2007

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**Selected Publications:**

1. C.-H. Huang, D. A. Peled, S. Schewe, *F. Wang* (corresponding author). A Game-Theoretic Foundation for the Maximum Software Resilience against Dense Errors, to appear in IEEE Transactions on Software Engineering (TSE).
2. *F. Wang*, S. Schewe, C.-H. Huang. An Extension of ATL with Strategy Interaction. ACM Transactions on Programming Languages and Systems (TOPLAS), Vol. 37, Issue 3, June 2015, ACM New York, NY, USA.
3. *F. Wang*, S. Schewe, J.-H. Wu. Complexity of Node Coverage Games. Theoretical Computer Science (TCS), Vol. 576, 20 April 2015, Pages 45–60, Elsevier.

4. F. Wang. Model-Checking Distributed Real-Time Systems with Propositions, Events, and Fairness Assumptions. To be archived in International Journal on Software Tools for Technology Transfer (STTT), Jan. 2014, Springer-Verlag.
5. F. Wang. Efficient Model-Checking of Dense-Time Systems with Time-Convexity Analysis. Theoretical Computer Science (TCS), Elsevier, Jan. 2013.
6. F. Wang, L.-W. Yao, Y.-L. Yang. Efficient Verification of Distributed Real-Time Systems with Broadcasting Behaviors. Real-Time Systems Journal, Vol. 47, Nr. 4, pp 285-318, July 2011.
7. F. Wang. Specification Formalisms and Models. Wiley Encyclopedia of Computer Science and Engineering 2008.
8. F. Wang, G.-D. Huang, F. Yu. *TCTL Inevitability Analysis of Dense-Time Systems: From Theory to Engineering*. IEEE Transactions on Software Engineering, Vol. 32, Nr. 7, July 2006, IEEE Computer Society.
9. F. Wang. *Symbolic Parametric Safety Analysis of Linear Hybrid Systems with BDD-like Data-Structures*. IEEE Transactions on Software Engineering, Vol. 31, Nr. 1, Jan. 2005. Also in CAV 2004, LNCS 3114, Springer-Verlag. (SCI)
10. F. Wang. *Inductive Composition of Numbers with Maximum, Minimum, and Addition - A New Theory for Program Execution-Time Analysis*. International Journal of Foundations of Computer Science, Vol. 15, Nr. 6, Dec. 2004, World Scientific. (SCI Extended)
11. F. Wang, H.-C. Yen. *Reachability Solution Characterization of Parametric Real-Time Systems*. Theoretical Computer Science 328 (2004), pp.187-201, Elsevier, special issue for CIAA 2003. (SCI)
12. F. Wang. *Formal Verification of Timed Systems -A Survey and Perspective*. Proceedings of the IEEE, Vol. 92, Nr. 8, August 2004, pp.1283-1307, IEEE. (SCI)
13. F. Wang, K. Schmidt, F. Yu, G.-D. Huang, B.-Y. Wang. *BDD-based Safety-Analysis of Concurrent Software with Pointer Data Structures using Graph Automorphism Symmetry Reduction*. IEEE Transactions on Software Engineering, Vol. 30, Nr. 6, June 2004, ISSN 0098-5589, pp.403-417, IEEE. (SCI)
14. F. Wang. *Efficient Verification of Timed Automata with BDD-like Data-Structures*. Invited to the International Journal on Software Tools for Technology Transfer (STTT), Vol. 6, Nr. 1, July 2004, Springer-Verlag, special issue for the 4<sup>th</sup> International Conference on Verification, Model-Checking, and Abstract Interpretation (VMCAI'2003), New York City, USA, January 2003.
15. F. Wang, G.-D. Hwang, F. Yu. *Symbolic Simulation of Industrial Real-Time and Embedded Systems -Experiments with the Bluetooth baseband communication protocol*. Invited to the first issue of the journal of Embedded Computing (JEC), Cambridge International Science Publishing
16. F. Wang, P.-A. Hsiung. *Efficient and User-Friendly Verification*. IEEE Transactions on Computers, Jan. 2002, Vol. 51, Nr. 1. (SCI)
17. F. Wang. *Parametric Analysis of Computer Systems*. Journal of Formal Methods in System Design, 17, 39-60, 2000, Kluwer Academic Publishers. (SCI extended)
18. F. Wang, C.-T. Lo. *Procedure-Level verification of Real-Time Concurrent Systems*. Journal of Real-Time Systems, Vol. 16, Nr. 1, Jan. 1999, pp.81-114, Kluwer Academic Publishers. (SCI)
19. J. Yang, F. Wang, A.K. Mok. *Symbolic Model-Checking for Event-Driven Real-Time Systems*. ACM TOPLAS (Transactions on Programming Languages and Systems), Vol. 19, No.2, March 1997, pp.386-412. (SCI)
20. F. Wang. *A Temporal Logic for Real-Time Partial-Ordering with Named Transactions*. Theoretical Computer Science, Vol. 181, Nr. 1, 15 July 1997, pp. 195-225; Guest Editor : R. Baeza Yates, E. Coles. A preliminary version also appears in Proceedings of Latin American Theoretical INformatics Symposium,

Santiago, Chile, April, 1995. LNCS 911, Springer-Verlag.

21. *F. Wang*. Parametric Timing Analysis of Real-Time Systems. Information and Computation, Vol. 130, Nr 2, Academic Press, ISSN 0890-5401, Nov. 1996; pp.131-150.
22. *F. Wang*, A.K. Mok, E.A. Emerson. Real-Time Distributed System Specification and Verification in APTL. ACM Transactions on Software Engineering and Methodology, Vol. 2, No. 4, October 1993, pp. 346-378.